

BHASWAR B. BHATTACHARYA

CONTACT INFORMATION

Department of Statistics and Data Science
The Wharton School
University of Pennsylvania
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Philadelphia, PA 19104
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RESEARCH INTERESTS

Nonparametric Statistics, Inference on Networks, Combinatorial Probability, Statistical Learning, Discrete and Computational Geometry.

EMPLOYMENT

- **Professor**, Department of Statistics and Data Science, The Wharton School, University of Pennsylvania, July 2026-present.
- **Associate Professor**, Department of Statistics and Data Science, The Wharton School, University of Pennsylvania, July 2022-June 2026.
- **Assistant Professor**, Department of Statistics and Data Science, The Wharton School, University of Pennsylvania, July 2016-June 2022.
- **Research Intern, Microsoft Research, Redmond**, June 2014-September 2014.

EDUCATION

- **Doctor of Philosophy (Ph.D.)**, Department of Statistics, Stanford University, September 2011-June 2016.
Thesis Title: Power of Graph-Based Two-Sample Tests
Advisor: Prof. Persi Diaconis
- **Master of Statistics (M. Stat.)**, Indian Statistical Institute, July 2009-May 2011.
- **Bachelor of Statistics (B. Stat.)**, Indian Statistical Institute, July 2006-May 2009.

AWARDS AND HONORS

1. Bernoulli Journal Lecture, *ISI World Statistics Congress*, 2025.
2. Special Invited Lecture, *Indian International Statistical Association (IISA) Annual Conference*, 2024.
3. Alfred P. Sloan Fellowship in Mathematics, 2021.
4. NSF CAREER Award, 2021.

5. *Probability Dissertation Award* from the Department of Statistics, Stanford University in 2016.
6. *Sabyasachi Roy Memorial Gold Medal* for the best Master's thesis in the 2009-2011 M. Stat. program of the Indian Statistical Institute, Kolkata, India.
7. Fellow of the *Kishore Vaigyanik Protsahan Yojana* (KVPY), 2008.

EDITORIAL BOARDS

- Associate Editor, *Annals of Statistics* (January 2025–)
- Associate Editor, *Journal of the American Statistical Association* (January 2023–)
- Associate Editor, *Annals of Applied Probability* (January 2022–)
- Associate Editor, *Bernoulli* (January 2022–)
- Associate Editor, *Sankhya, Series B* (January 2022–)

PREPRINTS

1. **Bhaswar B. Bhattacharya**, Sanchayan Bhowal, and Atmadeep Sengupta, Thresholds and fluctuations for colorful arithmetic progressions in sparse random colorings, [arXiv:2609.15086](#), 2026.
2. **Bhaswar B. Bhattacharya**, Sandip Das, Sk Samim Islam, Aashirwad Mohapatra, and Saumya Sen, Almost empty monochromatic triangles with many colors, [arXiv:2609.12325](#), 2026.
3. **Bhaswar B. Bhattacharya**, Pierfrancesco Dionigi, Ankan Ganguly, Giulio Zucal, Colorful exponential random graph models, [arXiv:2608.31130](#), 2026.
4. Anirban Chatterjee, Ziang Niu, **Bhaswar B. Bhattacharya**, Conditional mean independence and global sensitivity analysis using nearest neighbor graphs, [arXiv:2607.04692](#), 2026.
5. Somabha Mukherjee, Sanchayan Bhowal, Anirban Chatterjee, **Bhaswar B. Bhattacharya**, Ising models on inhomogeneous random graphs: Inference, local asymptotic minimaxity, and limit of experiments, [arXiv:2606.07065](#), 2026.
6. Sayak Chatterjee and **Bhaswar B. Bhattacharya**, Transitivity in inhomogeneous random tournaments, [arXiv:2606.02340](#), 2026.
7. **Bhaswar B. Bhattacharya**, Sandip Das, Sk Samim Islam, Aashirwad Mohapatra, Ishan Paul, and Saumya Sen, On the number of almost empty monochromatic triangles, [arXiv:2601.18951](#), 2026.
8. Sayak Chatterjee, Anirban Chatterjee, Abhinav Chakraborty, and **Bhaswar B. Bhattacharya**, Asymptotic normality of subgraph counts in sparse inhomogeneous random graphs, [arXiv:2512.12937](#), 2025.
9. Ankan Ganguly and **Bhaswar B. Bhattacharya**, Multiplexons: Limits of multiplex networks, [arXiv:2510.08639](#), 2025.
10. Anirban Chatterjee, Ziang Niu, and **Bhaswar B. Bhattacharya**, A kernel-based conditional two-sample test using nearest neighbors (with applications to calibration, regression curves, and simulation-based inference), [arXiv:2407.16550](#), 2024.
11. Ziang Niu and **Bhaswar B. Bhattacharya**, Distribution-free joint independence testing and robust independent component analysis using optimal transport, [arXiv:2211.15639](#), 2022, *under revision in the Journal of the American Statistical Association*.

PUBLICATIONS

12. **Bhaswar B. Bhattacharya**, Sanchayan Bhowal, Karambir Das, Laura Eslava, and Shaibal Karmakar, Thresholds and fluctuations of submultiplexes in random multiplex networks, *Canadian Mathematical Bulletin*, to appear, 2026.
13. Mauricio Daros and **Bhaswar B Bhattacharya**, Monochromatic subgraphs in randomly colored dense multiplex networks, *Advances in Applied Probability*, to appear, 2026.
14. Yangxinyu Xie and **Bhaswar B Bhattacharya**, Joint Poisson convergence of monochromatic hyperedges in multiplex hypergraphs, *Advances in Applied Mathematics*, Vol. 181, 103151, 2026.
15. Anirban Chatterjee, Soham Dan, and **Bhaswar B. Bhattacharya**, Higher-order graphon theory: fluctuations, degeneracies, and inference, *Annals of Statistics*, Vol. 54 (4), 2136–2163, 2026.
16. Bilol Banerjee, **Bhaswar B. Bhattacharya**, and Anil K. Ghosh, A ball divergence based measure for conditional independence testing, *Biometrika*, Volume 113 (3), asag027, 2026.
17. Nabarun Deb, **Bhaswar B. Bhattacharya**, and Bodhisattva Sen, Pitman efficiency lower bounds for multivariate distribution-free tests based on optimal transport, *Journal of the Royal Statistical Society, Series B*, Vol. 88 (3), 930–957, 2026.
18. **Bhaswar B. Bhattacharya**, Sandip Das, Sk Samim Islam, and Saumya Sen, Growth rates of the number of empty triangles and simplices, *Computational Geometry: Theory and Applications*, Vol. 130, 102197, 2026.
19. Somabha Mukherjee, Tianyu Liu, and **Bhaswar B. Bhattacharya**, Moderate deviation and Berry-Esseen bounds in the p -Spin Curie-Weiss model, *Indian Journal of Pure and Applied Mathematics (special issue in honor of K. R. Parthasarathy)*, to appear, 2025.
20. Anirban Chatterjee and **Bhaswar B. Bhattacharya**, Boosting the power of kernel two-sample tests, *Biometrika*, Vol. 112 (1), asae048, 2025.
21. Somabha Mukherjee, Jaesung Son, and **Bhaswar B. Bhattacharya**, Phase transitions of the maximum likelihood estimates in the p -spin Curie-Weiss model, *Bernoulli*, Vol. 31 (2), 1502–1526, 2025.
22. Trambak Banerjee, **Bhaswar B. Bhattacharya**, and Gourab Mukherjee, Bootstrapped edge count tests for nonparametric two-sample inference under heterogeneity, *Journal of Computational and Graphical Statistics*, Vol. 34 (1), 306–317, 2025.
23. Sherrie Xie, Maria Rieders, Srisa Changolkar, **Bhaswar B. Bhattacharya**, Elvis W Diaz, Michael Z Levy and Ricardo Castillo-Neyra, Enhancing mass vaccination programs with queueing theory and spatial optimization, *Frontiers in Public Health, Section on Infectious Diseases: Epidemiology and Prevention*, Vol. 12, 2024. (<https://doi.org/10.3389/fpubh.2024.1440673>)
24. Somabha Mukherjee, Ziang Niu, Sagnik Halder, **Bhaswar B. Bhattacharya**, and George Michailidis, High dimensional logistic regression under network dependence, *Journal of Machine Learning Research*, Vol. 25, 1–62, 2024.
25. Ricardo Castillo-Neyra, Sherrie Xie, Brinkley Raynor Bellotti, Elvis W. Diaz, Aris Saxena, Amparo M. Toledo, Gian Franco Condori-Luna, Maria Rieders, **Bhaswar B. Bhattacharya**, and Michael Z. Levy, Optimizing the location of vaccination sites to stop a zoonotic epidemic, *Scientific Reports*, Vol. 14, 15910, 2024.

26. **Bhaswar B. Bhattacharya**, Sayan Das, Somabha Mukherjee, and Sumit Mukherjee, Fluctuations of quadratic chaos, *Communications in Mathematical Physics*, Vol. 405, 237, 2024.
27. **Bhaswar B. Bhattacharya** and Rajarshi Mukherjee, Sparse uniformity testing, *IEEE Transactions on Information Theory*, Vol. 70 (9), 6371–6390, 2024.
28. Sagnik Nandy and **Bhaswar B. Bhattacharya**, Degree heterogeneity in higher-order networks: Inference in the hypergraph β -model, *IEEE Transactions on Information Theory*, Vol. 70 (8), 6000–6024, 2024.
29. **Bhaswar B. Bhattacharya**, Anirban Chatterjee, and Svante Janson, Fluctuations of subgraph counts in graphon based random graphs, *Combinatorics, Probability, and Computing*, Vol. 32 (3), 428–464, 2023.
30. Kwonsang Lee, **Bhaswar B. Bhattacharya**, Jing Qin, and Dylan S. Small, A nonparametric binomial likelihood approach for causal inference in instrumental variable models, *Journal of the Korean Statistical Society*, Vol. 52, 1055–1077, 2023.
31. Sayak Chatterjee, Dibyendu Saha, Soham Dan, and **Bhaswar B. Bhattacharya**, Two-sample tests for inhomogeneous random graphs in L_r norm: Optimality and asymptotics, *International Conference on Artificial Intelligence and Statistics (AISTATS)*, PMLR 206, 6903–6911, 2023.
32. Somabha Mukherjee, Jaesung Son, and **Bhaswar B. Bhattacharya**, Estimation in tensor Ising models, *Information and Inference: A Journal of the IMA*, Volume 11 (4), 1457–1500, 2022.
33. **Bhaswar B. Bhattacharya**, Sayan Das, and Sumit Mukherjee, Motif estimation via subgraph sampling: the fourth moment phenomenon, *Annals of Statistics*, Vol. 50 (2), 987–1011, 2022.
34. Somabha Mukherjee, Divyansh Agarwal, Nancy R. Zhang, and **Bhaswar B. Bhattacharya**, Distribution-free multisample test based on optimal matching with applications to single cell genomics, *Journal of the American Statistical Association*, Vol. 117 (538), 627–638, 2022.
35. **Bhaswar B. Bhattacharya**, Xiao Fang, and Han Yan, Normal approximation and fourth Moment theorems for monochromatic triangles, *Random Structures and Algorithms*, Vol. 60 (1), 25–53, 2022.
36. Aritra Banik, **Bhaswar B. Bhattacharya**, Sujoy Bhore, and Leonardo Martínez-Sandoval, Geometric systems of unbiased representatives, *Information Processing Letters*, Vol. 176, 106232, 2022.
37. Somabha Mukherjee, Jaesung Son, and **Bhaswar B. Bhattacharya**, Fluctuations of the magnetization in the p -spin Curie-Weiss model, *Communications in Mathematical Physics*, Vol. 387, 681–728, 2021.
38. **Bhaswar B. Bhattacharya** and Kavita Ramanan, Parameter estimation in undirected graphical models with hard constraints, *IEEE Transactions on Information Theory*, Vol. 67 (10), 6790–6809, 2021.
39. **Bhaswar B. Bhattacharya**, Somabha Mukherjee, and Sumit Mukherjee, Asymptotic distribution of Bernoulli quadratic forms, *Annals of Applied Probability*, Vol. 31 (4), 1548–1597, 2021.
40. **Bhaswar B. Bhattacharya**, Soham Bhattacharya, and Shirshendu Ganguly, Spectral edge in sparse random graphs: Upper and lower tail large deviations, *Annals of Probability*, Vol. 49 (4), 1847–1885, 2021.

41. Trambak Banerjee, **Bhaswar B. Bhattacharya**, and Gourab Mukherjee, A nearest-neighbor based nonparametric test for viral remodeling in heterogeneous single-cell proteomic data, *Annals of Applied Statistics*, Vol. 14 (4), 1777–1805, 2020.
42. **Bhaswar B. Bhattacharya**, Asymptotic distribution and detection thresholds for two-sample tests based on geometric graphs, *Annals of Statistics*, Vol. 48 (5), 2879–2903, 2020.
43. Soham Dan and **Bhaswar B. Bhattacharya**, Goodness-of-fit tests for inhomogeneous random graphs, *International Conference on Machine Learning (ICML)*, PMLR 119, 2335–2344, 2020.
44. Somabha Mukherjee and **Bhaswar B. Bhattacharya**, Replica symmetry in upper tails of mean-field hypergraphs, *Advances in Applied Mathematics*, Vol. 119, 102047, 2020.
45. **Bhaswar B. Bhattacharya** and Shirshendu Ganguly, Upper tails for edge eigenvalues of random graphs, *SIAM Journal on Discrete Mathematics*, Vol. 34 (2), 1069–1083, 2020.
46. **Bhaswar B. Bhattacharya**, Somabha Mukherjee, and Sumit Mukherjee, The second moment phenomenon for monochromatic subgraphs, *SIAM Journal on Discrete Mathematics*, Vol. 34 (1), 794–824, 2020.
47. **Bhaswar B. Bhattacharya**, Shirshendu Ganguly, Xuancheng Shao, and Yufei Zhao, Upper tail large deviations for arithmetic progressions in a random set, *International Mathematics Research Notices*, Vol. 2020 (1), 167–213, 2020.
48. **Bhaswar B. Bhattacharya** and Sumit Mukherjee, Limit theorems for monochromatic stars, *Random Structures and Algorithms*, Vol. 55 (4), 831–853, 2019.
49. Manjari Pradhan, **Bhaswar B. Bhattacharya**, Krishnendu Chakrabarty, and Bhargab B. Bhattacharya, Predicting X -sensitivity of circuit-inputs on test-coverage: A machine-learning approach, *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, Vol. 38 (12), 2343–2356, 2019.
50. Qingyuan Zhao, Dylan S. Small, and **Bhaswar B. Bhattacharya**, Sensitivity analysis for inverse probability weighting estimators via the percentile bootstrap, *Journal of the Royal Statistical Society, Series B*, Vol. 81 (4), 735–761, 2019.
51. **Bhaswar B. Bhattacharya**, A general asymptotic framework for distribution-free graph-based two-sample tests, *Journal of the Royal Statistical Society, Series B*, Vol. 81 (3), 575–602, 2019.
52. **Bhaswar B. Bhattacharya** and Sumit Mukherjee, Monochromatic subgraphs in randomly colored graphons, *European Journal of Combinatorics*, Vol. 81, 328–353, 2019.
53. Aritra Banik, **Bhaswar B. Bhattacharya**, Sandip Das, and Sreeja Das, The 1-dimensional discrete Voronoi game, *Operations Research Letters*, Vol. 47, 115–121, 2019.
54. **Bhaswar B. Bhattacharya** and Sumit Mukherjee, Inference in Ising models, *Bernoulli*, Vol. 24 (1), 493–525, 2018.
55. **Bhaswar B. Bhattacharya**, Shirshendu Ganguly, Eyal Lubetzky, and Yufei Zhao, Upper tails and independence polynomials in random graphs, *Advances in Mathematics*, Vol. 319, 313–347, 2017.
56. Aritra Banik, **Bhaswar B. Bhattacharya**, Sandip Das, and Satyaki Mukherjee, The discrete Voronoi game in $\mathbb{R}^2$, *Computational Geometry: Theory and Applications*, Vol. 63, 53–62, 2017.
57. **Bhaswar B. Bhattacharya** and Sumit Mukherjee, Degree sequence of random permutation graphs, *Annals of Applied Probability*, Vol. 27 (1), 439–484, 2017.

58. **Bhaswar B. Bhattacharya**, Persi Diaconis, and Sumit Mukherjee, Universal poisson and normal limit theorems in graph coloring problems with connections to extremal combinatorics, *Annals of Applied Probability*, Vol. 27 (1), 337–394, 2017.
59. **Bhaswar B. Bhattacharya**, Collision times in multicolor urn models and sequential graph coloring with applications to discrete logarithms, *Annals of Applied Probability*, Vol. 26 (6), 3286–3318, 2016.
60. **Bhaswar B. Bhattacharya** and Subhabrata Sen, High temperature asymptotics of orthogonal mean-field spin glasses, *Journal of Statistical Physics*, Vol. 162 (1), 63–80, 2016.
61. Deepan Basu, Kinjal Basu, **Bhaswar B. Bhattacharya**, and Sandip Das, Almost empty monochromatic triangles in planar point sets, *Discrete Applied Mathematics*, Vol. 210, 207–213, 2016.
62. **Bhaswar B. Bhattacharya** and Gregory Valiant, Testing closeness with unequal sized samples, *Neural Information Processing System (NIPS)*, 2611–2619, 2015.
63. **Bhaswar B. Bhattacharya** and Sumit Mukherjee, Exact and asymptotic results on coarse Ricci curvature of graphs, *Discrete Mathematics*, Vol. 338, 23–42, 2015.
64. **Bhaswar B. Bhattacharya**, Sayantan Das, and Shirshendu Ganguly, Minimum-weight edge discriminator in hypergraphs, *Electronic Journal of Combinatorics*, Vol. 21 (3), #P3.18, 2014.
65. Aritra Banik, **Bhaswar B. Bhattacharya**, and Sandip Das, The minimum enclosing circle of a set of fixed points and a mobile point, *Computational Geometry: Theory and Applications*, Vol. 47 (9), 891–898, 2014.
66. Aritra Banik, **Bhaswar B. Bhattacharya**, and Sandip Das, Optimal strategies for the one-round discrete Voronoi game on a line, *Journal of Combinatorial Optimization*, Vol. 26, 655–669, 2013.
67. **Bhaswar B. Bhattacharya** and Sandip Das, On pseudo-convex partitions of a planar point set, *Discrete Mathematics*, Vol. 313 (21), 2401–2408, 2013.
68. **Bhaswar B. Bhattacharya** and Subhas C. Nandy, New variations of the maximum coverage facility location problem, *European Journal of Operational Research*, Vol. 224, 477–485, 2013.
69. **Bhaswar B. Bhattacharya** and Sandip Das, Disjoint empty convex pentagons in planar point sets, *Periodica Mathematica Hungarica*, Vol. 66 (1), 73–86, 2013.
70. Riddhipratim Basu, **Bhaswar B. Bhattacharya**, and Tanmoy Talukdar, The projection median of a set of points in $\mathbb{R}^d$, *Discrete and Computational Geometry*, Vol. 47 (2), 329–346, 2012.
71. **Bhaswar B. Bhattacharya** and Sandip Das, Holes or empty-pseudo-triangles in planar point sets, *Moscow Journal of Combinatorics and Number Theory*, Vol. 2 (1), 16–46, 2012.
72. **Bhaswar B. Bhattacharya** and Sandip Das, On the minimum size of a point set containing a 4-Hole and a disjoint 5-Hole, *Studia Scientiarum Mathematicarum Hungarica*, Vol. 48 (4), 445–457, 2011.
73. **Bhaswar B. Bhattacharya**, On the Fermat-Weber point of a polygonal chain and its generalizations, *Fundamenta Informaticae*, Vol. 107 (4), 331–343, 2011.
74. **Bhaswar B. Bhattacharya**, Maximizing Voronoi regions of a set of points enclosed in a circle with applications to facility location, *Journal of Mathematical Modelling and Algorithms*, Vol. 9 (4), 375–392, 2010.

75. **Bhaswar B. Bhattacharya** and Sandip Das, Geometric proof of a Ramsey-type result for disjoint empty convex polygons I and II, *Geombinatorics*, Vol. XX (1), 5–16, and Vol. XIX (4), 146–155, 2010.

TEACHING

- Fall 2025 (Wharton): Mathematical Statistics (STAT 4320), Mathematical Statistics (STAT 9700)
- Fall 2024 (Wharton): Mathematical Statistics (STAT 4320), Mathematical Statistics (STAT 9700)
- Fall 2023 (Wharton): Mathematical Statistics (STAT 4320), Mathematical Statistics (STAT 9700)
- Spring 2023 (Wharton): Mathematical Statistics (STAT 9700)
- Fall 2022 (Indian Statistical Institute): Combinatorial Statistics and Network Analysis
- Spring 2022 (Wharton): Mathematical Statistics (STAT 432)
- Fall 2021 (Wharton): Mathematical Statistics (STAT 970)
- Spring 2021 (Wharton): Seminar in Advanced Applications in Statistics (STAT 991)
- Fall 2020 (Wharton): Mathematical Statistics (STAT 970)
- Fall 2019 (Wharton): Mathematical Statistics (STAT 970)
- Fall 2018 (Wharton): Statistical Inference (STAT 431/511), Mathematical Statistics (STAT 970)
- Fall 2017 (Wharton): Statistical Inference (STAT 431/511), Mathematical Statistics (STAT 970)
- Spring 2017 (Wharton): Seminar in Advanced Applications in Statistics (STAT 991)
- Fall 2016 (Wharton): Statistical Inference (STAT 431/511)
- Summer 2013 (Stanford): Qualifying Exams Workshop (STATS 302)

SOFTWARE

- Invited contribution to the **Wolfram Demonstration Project**: *Fermat-Weber Point of a Polygonal Chain*.

ADVISING

- Ph.D. Dissertation Advisor:
 1. Ziang Niu (co-advisor, jointly with Eugene Katsevich), Department of Statistics and Data Science, The Wharton School, University of Pennsylvania (2023–).
 2. Sayak Chatterjee, Department of Statistics and Data Science, The Wharton School, University of Pennsylvania, (2022–).
 3. Anirban Chatterjee, Department of Statistics and Data Science, The Wharton School, University of Pennsylvania, 2025.
 - *Current Position*: Kruskal Instructor, Department of Statistics, University of Chicago during 2025-2026. Starting as Assistant Professor, Department of Mathematics and Statistics, Boston University from Fall 2026.

4. Mauricio Daros Andrade, Department of Statistics and Data Science, The Wharton School, University of Pennsylvania, 2025.
– *Current Position*: Postdoctoral Researcher, FGV EMap, Brazil.
5. Sagnik Nandy (co-advisor, primary advisor Zongming Ma), Department of Statistics and Data Science, The Wharton School, University of Pennsylvania, Ph.D. 2024.
– *Current Position*: Assistant Professor, Department of Statistics, Ohio State University.
6. Somabha Mukherjee, Department of Statistics and Data Science, The Wharton School, University of Pennsylvania, Ph.D. 2021.
– *Current Position*: Assistant Professor, Department of Statistics and Data Science, National University of Singapore.

- Postdoc Mentor: Ankan Ganguly (Spring 2023).

GRANT SUPPORT

- Alfred P. Sloan Fellowship in Mathematics, 2021–2023.
- Wharton Dean’s Research Fund, 2021–2023.
- NSF CAREER DMS 2046393 (Principal Investigator): *Geometric and combinatorial methods for distribution-free inference and dependent network data*, 07/01/2021–06/30/2026.
- NSF DMS 2113771 (Co-Principal Investigator with Debashis Mondal): *Distance based analysis for complex high-dimensional data*, 07/01/2021–06/30/2024.
- NIH/NIAID R01-AI168291 (Co-Investigator): *Spatially responsive mass vaccination strategies for urban rabies*, 04/08/22–03/31/27 (Principal Investigator: Ricardo Castillo-Neyra).
- NIH-R01AI146129 (Co-Investigator): *An immune system for the city: a new paradigm for control of urban disease vectors*, 07/01/2019–06/30/2024 (Principal Investigators: Michael Z. Levy and Valerie A. Paz-Soldan).